HW 5 — $TM_{1,1}$ in a Rectangular Guide & Jackson 8.2(a)

black = write it down · **red = the answer** · *teal = where it came from / my note* · grey = optional

This one is almost entirely lecture-backed. Lec 11, s10–11 is titled "TM in Rectangular Waveguide?" and shows this exact figure — which is where the question comes from. He has since filled those slides in, and they agree with the route below. Lec 10, s17 gives the TM equations, Lec 11, s13–17 and Lec 12 give the energy flow, and Lec 12, s14–16 works the same power calculation for **TE**. Q1 is the TM twin of that worked example.

Question 1 asks for two things:

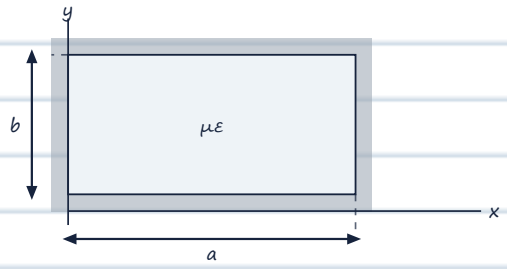
derive the **fields** of $TM_{1,1}$ — **Steps 1–4**

calculate the total power flow $\langle P \rangle$ — **Steps 5–6**

Everything below was checked numerically: the fields satisfy $E_z = 0$ and tangential $\mathbf{E} = 0$ on all four walls (to 10^{-16}), satisfy $\nabla\times\mathbf{E} = i\omega\mu\mathbf{H}$ exactly, satisfy the Helmholtz equation (residual → 0 as h^2), and the power integral was done numerically and agrees with the closed form to 16 digits.

Question 1 — $TM_{1,1}$ in a rectangular waveguide

For the waveguide of Jackson Fig 8.5 (a hollow rectangular pipe, cross-section a × b, filled with μ, ε), **derive the fields of the $TM_{1,1}$ mode** and **calculate its total power flow ⟨P⟩**.



Jackson Fig 8.5, as reproduced on Lec 11, s10. *The wave travels along* **z**, *out of the page.*

ANSWER

Fields (all × $e^{i(kz-\omega t)}$):

$$E_z = E_0 \sin\left(\frac{\pi x}{a}\right) \sin\left(\frac{\pi y}{b}\right)$$

$$E_x = \frac{ik}{\gamma^2}\,\frac{\pi}{a} E_0 \cos\frac{\pi x}{a} \sin\frac{\pi y}{b}, \quad E_y = \frac{ik}{\gamma^2}\,\frac{\pi}{b} E_0 \sin\frac{\pi x}{a} \cos\frac{\pi y}{b}$$

$$H_x = -\frac{i\varepsilon\omega}{\gamma^2}\,\frac{\pi}{b} E_0 \sin\frac{\pi x}{a} \cos\frac{\pi y}{b}, \quad H_y = +\frac{i\varepsilon\omega}{\gamma^2}\,\frac{\pi}{a} E_0 \cos\frac{\pi x}{a} \sin\frac{\pi y}{b}, \quad H_z = 0$$

$$\gamma^2 = \pi^2(1/a^2 + 1/b^2) = \mu\varepsilon\omega_\lambda^2, \quad k = \sqrt{\mu\varepsilon\omega^2 - \gamma^2}$$

Power:

$$\langle P\rangle = \frac{\varepsilon\omega k\, ab\, E_0^2}{8\gamma^2} = \frac{1}{2\sqrt{\mu\varepsilon}}\,\frac{\omega^2}{\omega_\lambda^2}\left(1 - \frac{\omega_\lambda^2}{\omega^2}\right)^{1/2} \varepsilon E_0^2\,\frac{ab}{4}$$

Flow: solve the 2D equation for E_z → the walls fix the allowed π/a, π/b → set m = n = 1 → get $\mathbf{E}_t$ and $\mathbf{H}_t$ from E_z → integrate the Poynting vector.

Step 0 — Where everything starts

The **only** inputs are Maxwell's two curl equations, with every field carrying $e^{i(kz-\omega t)}$:

$$\nabla\times\vec{E} = i\omega\mu\vec{H}, \qquad \nabla\times\vec{H} = -i\omega\varepsilon\vec{E}$$

Because the z-dependence is e^{ikz}, every $\partial/\partial z$ becomes **ik**. So split the del operator, and split the fields into transverse + z parts:

$$\nabla = \nabla_t + ik\hat{z}, \qquad \vec{E} = \vec{E}_t + \hat{z}E_z, \qquad \vec{H} = \vec{H}_t + \hat{z}H_z$$

TM mode means $H_z = 0$, $E_z \neq 0$

Step 1 — Build $\vec{E}_t$ and $\vec{H}_t$ out of E_z

Put the splits into the **second** curl equation. With $H_z = 0$:

$$(\nabla_t + ik\hat{z})\times\vec{H}_t = -i\omega\varepsilon(\vec{E}_t + \hat{z}E_z)$$

Same four-way sort as above: $\nabla_t\times\vec{H}_t$ points along $\hat{z}$, while $ik\hat{z}\times\vec{H}_t$ is transverse. Keeping the **transverse** parts:

$$ik(\hat{z}\times\vec{H}_t) = -i\omega\varepsilon\vec{E}_t$$

Cross both sides with $\hat{z}$, using $\hat{z}\times(\hat{z}\times\vec{A}) = -\vec{A}$ for any transverse $\vec{A}$:

$$\vec{H}_t = \frac{\omega\varepsilon}{k}(\hat{z}\times\vec{E}_t) \qquad (1)$$

Now the **first** curl equation. Substitute both splits and multiply out:

$$(\nabla_t + ik\hat{z})\times(\vec{E}_t + \hat{z}E_z) = i\omega\mu(\vec{H}_t + \hat{z}H_z)$$

That is **four** terms on the left. Sort them by direction:

#	term	points where?
1	$\nabla_t\times\vec{E}_t$	both lie flat in the cross-section → their cross is along $\hat{z}$
2	$\nabla_t\times(\hat{z}E_z)$	**transverse**
3	$ik\hat{z}\times\vec{E}_t$	$\hat{z}$ crossed with a flat vector stays flat → **transverse**
4	$ik\hat{z}\times(\hat{z}E_z)$	$\hat{z}\times\hat{z} = 0$ → **gone**

The right side splits the same way: $i\omega\mu\vec{H}_t$ is transverse, $i\omega\mu\hat{z}H_z$ is along z. A transverse vector can never equal a z-vector, so the one equation breaks into **two independent ones**:

transverse: $\nabla_t E_z\times\hat{z} + ik(\hat{z}\times\vec{E}_t) = i\omega\mu\vec{H}_t$ *terms 2 + 3 — the one we want*

z: $\nabla_t\times\vec{E}_t = i\omega\mu\hat{z}H_z$ *term 1 — not needed here*

Term 2 rewritten. *The product rule for curl is $\nabla\times(f\vec{A}) = (\nabla f)\times\vec{A} + f(\nabla\times\vec{A})$. Here $\vec{A} = \hat{z}$ is a constant vector, so $\nabla\times\hat{z} = 0$ and only the first piece survives: $\nabla_t\times(\hat{z}E_z) = \nabla_t E_z\times\hat{z}$.*

Substitute ① on the right:

$$\nabla_t E_z\times\hat{z} = i\left[\frac{\omega^2\mu\varepsilon}{k} - k\right](\hat{z}\times\vec{E}_t) = i\frac{\gamma^2}{k}(\hat{z}\times\vec{E}_t) \qquad \gamma^2 \equiv \mu\varepsilon\omega^2 - k^2$$

Write the left side as $-\hat{z}\times\nabla_t E_z$, then cross both sides with $\hat{z}$ again:

$$\nabla_t E_z = -i\frac{\gamma^2}{k}\vec{E}_t \quad\Rightarrow\quad \vec{E}_t = \frac{ik}{\gamma^2}\nabla_t E_z \qquad 1/(-i) = i$$

And back into ①:

$$\vec{H}_t = \frac{i\omega\varepsilon}{\gamma^2}(\hat{z}\times\nabla_t E_z)$$ matches Lec 10, s17

This is the key result. *Both transverse fields are now just derivatives of the single scalar E_z. Find E_z and the mode is finished.*

Step 2 — The equation E_z obeys

Every Cartesian field component satisfies the wave equation $\nabla^2 E_z + \mu\varepsilon\omega^2 E_z = 0$. Split the Laplacian, remembering $\partial^2/\partial z^2 \to (ik)^2 = -k^2$:

$$\nabla^2 = \nabla_t^2 + \frac{\partial^2}{\partial z^2} = \nabla_t^2 - k^2$$

$$\Rightarrow \quad \nabla_t^2 E_z + (\mu\varepsilon\omega^2 - k^2) E_z = 0 \quad \Rightarrow \quad \nabla_t^2 E_z + \gamma^2 E_z = 0$$

The wall runs parallel to z, so E_z is a **tangential** component there. On a perfect conductor tangential E vanishes:

$$E_z = 0 \text{ on all four walls}$$

Step 3 — Separate the variables

The cross-section is a rectangle, so try $E_z = X(x)Y(y)$:

$$X''Y + XY'' + \gamma^2 XY = 0 \qquad \nabla_t^2 = \partial^2/\partial x^2 + \partial^2/\partial y^2$$

Divide through by XY:

$$\frac{X''}{X} + \frac{Y''}{Y} + \gamma^2 = 0$$

The first term depends only on x, the second only on y, and they add to a constant. The only way that can hold for *all* x and y is if each is **separately** constant:

$$\frac{X''}{X} = -k_x^2, \qquad \frac{Y''}{Y} = -k_y^2, \qquad k_x^2 + k_y^2 = \gamma^2$$

Each is simple harmonic motion, so

$$X = A\cos k_x x + B\sin k_x x \qquad \text{and the same for } Y$$

Step 4 — The walls kill the cosines and quantise the sines

At **x = 0**: $\cos(0) = 1$ but $\sin(0) = 0$, so $E_z = 0$ forces **A = 0** — only the sine survives.

At **x = a**: now we need $\sin k_x a = 0$, and a sine vanishes only at whole multiples of π:

$$k_x a = m\pi \quad \Rightarrow \quad k_x = m\pi/a, \qquad m = 1, 2, 3, \ldots$$

Identical argument in y:

$$k_y = n\pi/b, \qquad \gamma^2 = \left(\frac{m\pi}{a}\right)^2 + \left(\frac{n\pi}{b}\right)^2$$ = $\mu\varepsilon\omega_\lambda^2$, Lec 12 s15

Why m and n both start at 1. If m = 0 then $\sin(0 \cdot \pi x/a) = 0$ everywhere, so $E_z \equiv 0$ and by Step 1 every other component dies too. So $TM_{1,0}$, $TM_{0,1}$, $TM_{0,0}$ do not exist, and the lowest TM mode is **$TM_{1,1}$**. (TE uses cosines instead, so $TE_{1,0}$ survives — that is his worked case.)

Step 5 — Set m = n = 1 and write every component

$$E_z = E_0 \sin\frac{\pi x}{a}\sin\frac{\pi y}{b}\,e^{i(kz-\omega t)}, \qquad \gamma^2 = \pi^2\left(\frac{1}{a^2}+\frac{1}{b^2}\right)$$

Take the transverse gradient — the only differentiation in the whole problem:

$$\nabla_t E_z = \frac{\pi}{a}E_0\cos\frac{\pi x}{a}\sin\frac{\pi y}{b}\,\hat{x} + \frac{\pi}{b}E_0\sin\frac{\pi x}{a}\cos\frac{\pi y}{b}\,\hat{y}$$

Now just feed that into the two Step 1 formulas. For $\vec{E}_t$, multiply by ik/γ^2:

$$E_x = \frac{ik\pi}{a\gamma^2}E_0\cos\frac{\pi x}{a}\sin\frac{\pi y}{b}, \qquad E_y = \frac{ik\pi}{b\gamma^2}E_0\sin\frac{\pi x}{a}\cos\frac{\pi y}{b}$$

For $\vec{H}_t$, multiply by $i\omega\varepsilon/\gamma^2$ **and** apply $\hat{z}\times$. Since $\hat{z}\times\hat{x} = \hat{y}$ and $\hat{z}\times\hat{y} = -\hat{x}$, that **swaps the components and flips one sign**:

$$H_x = -\frac{i\omega\varepsilon\pi}{b\gamma^2}E_0\sin\frac{\pi x}{a}\cos\frac{\pi y}{b}, \qquad H_y = +\frac{i\omega\varepsilon\pi}{a\gamma^2}E_0\cos\frac{\pi x}{a}\sin\frac{\pi y}{b}$$

$H_z = 0$ *by definition of TM*

Wall check. *At $x = 0$ and $x = a$ the tangential components are E_y and E_z, and both carry $\sin(\pi x/a) \to 0$ ✓. At $y = 0$ and $y = b$ they are E_x and E_z, both carrying $\sin(\pi y/b) \to 0$ ✓.*

Step 6 — The Poynting vector, derived

Write $\psi \equiv E_z$ from here on, to keep it short.

$$\langle \vec{S} \rangle = \frac{1}{2} Re(\vec{E} \times \vec{H}^*), \qquad \vec{E} = \vec{E}_t + \hat{z}\psi$$

The piece $\hat{z}\psi \times \vec{H}_t^*$ is **transverse** ($\hat{z}$ crossed with a transverse vector gives a transverse vector), so it has **no z-component** and cannot carry power along the guide. Only one term survives:

$$\langle S_z \rangle = \frac{1}{2} Re(\vec{E}_t \times \vec{H}_t^*) \cdot \hat{z}$$

Put in the Step 1 forms:

$$\vec{E}_t \times \vec{H}_t^* = \frac{ik}{\gamma^2} \nabla_t \psi \times \left[\frac{i\omega\varepsilon}{\gamma^2} (\hat{z} \times \nabla_t \psi) \right]^*$$

The conjugate flips the sign of i, so $(i)(-i) = +1$ and the prefactors multiply to a **real positive** number:

$$= \frac{k\omega\varepsilon}{\gamma^4} \nabla_t \psi \times (\hat{z} \times \nabla_t \psi^*)$$

Now use the triple-product identity $\vec{A} \times (\vec{B} \times \vec{C}) = \vec{B}(\vec{A} \cdot \vec{C}) - \vec{C}(\vec{A} \cdot \vec{B})$, with $\vec{A} = \nabla_t \psi$, $\vec{B} = \hat{z}$, $\vec{C} = \nabla_t \psi^*$:

$$\nabla_t \psi \times (\hat{z} \times \nabla_t \psi^*) = \hat{z}(\nabla_t \psi \cdot \nabla_t \psi^*) - \nabla_t \psi^* (\nabla_t \psi \cdot \hat{z})$$

The second term is **zero**: $\nabla_t \psi$ lies in the cross-section, so it is perpendicular to $\hat{z}$. What is left is

$$= \hat{z} \left| \nabla_t \psi \right|^2$$

$$\Rightarrow \langle S_z \rangle = \frac{\omega k}{2\gamma^4} \varepsilon \left| \nabla_t \psi \right|^2$$

*For a **TE** mode the same algebra runs with $\psi = H_z$ and gives μ in place of ε. He writes the two together as a brace $\{\varepsilon, \mu\}$.*

Step 7 — Integrate — Green's first identity

$$\langle P \rangle = \int_A \langle S_z \rangle \, da = \frac{\omega k}{2\gamma^4} \varepsilon \int_A (\nabla_t \psi)^* \cdot (\nabla_t \psi) \, da$$

That integral is done with **Green's first identity**, which trades a squared gradient for a boundary term plus a Laplacian:

$$\int_A (\nabla_t \psi)^* \cdot (\nabla_t \psi) \, da = \oint_S \psi^* \frac{\partial \psi}{\partial n} dL - \int_A \psi^* \nabla_t^2 \psi \, da$$

Both pieces now collapse:

① the boundary integral is **zero**, because $\psi = 0$ on the wall (TM) — or $\partial\psi/\partial n = 0$ (TE). *This is what the wall condition was for.*

② $\nabla_t^2\psi = -\gamma^2\psi$, straight from Step 2.

$\Rightarrow \int|\nabla_t\psi|^2\,da = \gamma^2\int\psi^*\psi\,da$ *one γ^2 comes out and cancels*

$$\langle P\rangle = \frac{\omega k}{2\gamma^2}\,\varepsilon\int_A\psi^*\psi\,da$$

Finally put $k = \sqrt{\mu\varepsilon}\sqrt{\omega^2-\omega_\lambda^2}$ and $\gamma^2 = \mu\varepsilon\omega_\lambda^2$:

$$\langle P\rangle = \frac{1}{2\sqrt{\mu\varepsilon}}\left(\frac{\omega}{\omega_\lambda}\right)^2\left(1-\frac{\omega_\lambda^2}{\omega^2}\right)^{1/2}\varepsilon\int\psi^*\psi\,da$$

General result — *true for any cross-section and any mode. Only the last integral knows the shape of the pipe.*

Step 8 — The one integral that is left

Put in $\psi = E_z$. The x and y integrals separate, and a squared sine over a whole number of half-periods averages to ½:

$$\int\psi^*\psi\,da = E_0^2\int_0^a\sin^2\frac{\pi x}{a}\,dx\int_0^b\sin^2\frac{\pi y}{b}\,dy = E_0^2\,\frac{a}{2}\,\frac{b}{2} = E_0^2\,\frac{ab}{4}$$

ANSWER — power flow

$$\langle P\rangle = \frac{1}{2\sqrt{\mu\varepsilon}}\left(\frac{\omega}{\omega_\lambda}\right)^2\left(1-\frac{\omega_\lambda^2}{\omega^2}\right)^{1/2}\varepsilon E_0^2\,\frac{ab}{4}$$

Equivalently $\langle P\rangle = \varepsilon\omega k\,ab\,E_0^2\,/\,8\gamma^2$.

Against his $TE_{1,0}$: *there $\psi = H_0\cos(\pi x/a)$, the brace picks μ, and $\int\psi^*\psi\,da = H_0^2(ab/2)$ — the $n = 0$ direction gives b, not b/2, because $\cos^2(0) = 1$ all the way across. That is the {ab/4 ; ab/2} brace on* Lec 12, s16*. Our $m = n = 1$ is the ab/4 branch.*

■ END OF QUESTION 1 ■

Question 2 — Jackson 8.2(a)

> A transmission line of two concentric metal cylinders (conductivity σ, skin depth δ), inner radius a, outer radius b, filled with a lossless dielectric (μ, ε), carrying a **TEM mode**. Show that the time-averaged power flow along the line is $P = \sqrt{\mu/\varepsilon}\ \pi a^2 |H_0|^2 \ln(b/a)$, where H_0 is the peak azimuthal magnetic field at the surface of the inner conductor.

ANSWER

$$P = \sqrt{\mu/\varepsilon}\ \pi a^2 \left|H_0\right|^2 \ln\left(\frac{b}{a}\right)$$

Flow: *impose $E_z = H_z = 0$ on Maxwell → the transverse field turns out to be an electrostatic one → solve it for two concentric circles → integrate the Poynting vector.*

Step 1 — What TEM forces

*TEM means **$E_z = 0$ and $H_z = 0$**. Feed that into the same split Maxwell equations used in Question 1.*

*Take the **z-components** first. From $\nabla\times\vec{E} = i\omega\mu\vec{H}$, and from $\nabla\cdot\vec{E} = 0$ (no free charge):*

$$\nabla_t\times\vec{E}_t = i\omega\mu H_z = 0 \quad\Rightarrow\quad \nabla_t\times\vec{E}_t = 0$$

$$\nabla_t\cdot\vec{E}_t + ikE_z = 0 \quad\Rightarrow\quad \nabla_t\cdot\vec{E}_t = 0$$

*Curl-free **and** divergence-free in two dimensions — those are exactly the conditions of a 2D **electrostatics** problem, even though this is a travelling wave.*

A curl-free field is a gradient, and then zero divergence makes it harmonic:

$$\vec{E}_t = -\nabla_t\phi, \qquad \nabla_t^2\phi = 0$$ cf. Lec 10, s13–15

Now the **transverse** parts. Each curl equation gives $\vec{H}_t$ in terms of $\vec{E}_t$:

$$ik(\hat{z}\times\vec{E}_t) = i\omega\mu\vec{H}_t \;\Rightarrow\; \vec{H}_t = \frac{k}{\omega\mu}(\hat{z}\times\vec{E}_t) \quad ①$$

$$ik(\hat{z}\times\vec{H}_t) = -i\omega\varepsilon\vec{E}_t \;\Rightarrow\; \vec{H}_t = \frac{\omega\varepsilon}{k}(\hat{z}\times\vec{E}_t) \quad ②$$

① and ② must agree, which fixes k:

$$\frac{k}{\omega\mu} = \frac{\omega\varepsilon}{k} \;\Rightarrow\; k^2 = \mu\varepsilon\omega^2 \;\Rightarrow\; k = \omega\sqrt{\mu\varepsilon}$$ so $\gamma^2 = \mu\varepsilon\omega^2 - k^2 = 0$ — TEM has no cutoff

Putting that k back into ② gives the ratio of the two fields:

$$\vec{H}_t = \sqrt{\varepsilon/\mu}(\hat{z}\times\vec{E}_t) = \frac{1}{Z}(\hat{z}\times\vec{E}_t), \qquad Z \equiv \sqrt{\mu/\varepsilon}$$

Step 2 — Solve it for two concentric circles

Each conductor is an equipotential, and the two are concentric, so ϕ can depend only on ρ. In polar coordinates Laplace's equation reads

$$\frac{1}{\rho}\frac{d}{d\rho}\left(\rho\frac{d\phi}{d\rho}\right) = 0 \;\Rightarrow\; \rho\frac{d\phi}{d\rho} = \text{constant} \;\Rightarrow\; \phi = A\ln\rho + B$$

$$E_\rho = -\frac{d\phi}{d\rho} = -\frac{A}{\rho}$$ everything $\propto 1/\rho$, like the field of a line charge

From Step 1, $\vec{H}_t = (1/Z)(\hat{z}\times\vec{E}_t)$, and $\hat{z}\times\hat{\rho} = \hat{\phi}$, so **H** is azimuthal with $H_\phi = E_\rho/Z$. Writing H_o for its value at $\rho = a$, as the question does:

$$H_\phi = H_o\frac{a}{\rho}, \qquad E_\rho = \sqrt{\mu/\varepsilon}\,H_o\frac{a}{\rho}$$

Step 3 — The Poynting vector

E is radial and **H** azimuthal, and $\hat{\rho}\times\hat{\phi} = \hat{z}$, so the flow is straight along the line:

$$\langle S_z\rangle = \frac{1}{2}\,\mathrm{Re}(E_\rho H_\phi^*) = \frac{1}{2}\sqrt{\mu/\varepsilon}\,|H_o|^2\,\frac{a^2}{\rho^2}$$

Step 4 — Integrate over the annulus

The dielectric fills $a \le \rho \le b$. A ring of radius ρ has area $2\pi\rho\,d\rho$:

$$P = \int_a^b \frac{1}{2}\sqrt{\mu/\varepsilon}\,|H_0|^2\,\frac{a^2}{\rho^2}\,2\pi\rho\,d\rho$$

The ρ^2 downstairs cancels one ρ upstairs, leaving $d\rho/\rho$:

$$= \pi\sqrt{\mu/\varepsilon}\,|H_0|^2 a^2 \int_a^b \frac{d\rho}{\rho}$$

$$\int_a^b \frac{d\rho}{\rho} = \ln\rho\,\Big|_a^b = \ln b - \ln a = \ln\left(\frac{b}{a}\right)$$

ANSWER — 8.2(a)

$$P = \sqrt{\mu/\varepsilon}\,\pi a^2 |H_0|^2 \ln\left(\frac{b}{a}\right) \quad \checkmark$$

*The **ln(b/a)** is just $\int d\rho/\rho$ — nothing deeper. It is the signature of a $1/\rho$ field, and it shows up again in the impedance and inductance of the same line.*

■ END OF QUESTION 2 ■

OPTIONAL **The two questions are the same calculation.** Both are $\tfrac{1}{2}\mathrm{Re}(\mathbf{E}\times\mathbf{H}^*)\cdot\hat{z}$ integrated over the cross-section. Q1 gives $ab/4$ because the integrand is $\sin^2\times\cos^2$ over a rectangle; Q2 gives $\ln(b/a)$ because the integrand is $1/\rho^2$ over an annulus. Only the shape of the pipe is different.